

Q1.

Suppose there exists a pair of relations $R_1(X, Y, Z)$ and $R_2(Y, Z, W)$ having $t_1 > 0$ and $t_2 > 0$ tuples, respectively. Consider that X, Y, Z and W take integer values only. Without making any further assumptions, find out the minimum and maximum possible number of tuples that may appear in the resulting relations provided by the following operations.

i) $\pi_{Y, Z}(R_1 \bowtie R_2)$.

• MIN : Zero.

We will get no tuples from the above query if we consider the case where the common tuples of R_1 and R_2 i.e. Y, Z contains no common values w.r.t which we can perform the natural join i.e. $R_1(X, Y, Z) \cap R_2(Y, Z, W) = \emptyset$.

As, natural join is defined as $R_1 \bowtie R_2 = R_1 \times_{R_1.A = R_2.A \wedge \dots \wedge R_1.N = R_2.N} R_2$.

Insufficiency in common attribute will ~~not~~ lead to ZERO tuples even before ~~project~~ selecting Y, Z .

• MAX: $\min(t_1, t_2)$.

To get the maximum no. of tuples we need ~~get~~ to have a matching pair in the other schema for every common attribute. By doing this we maximized our cartesian product. Then we are selecting the common attribute Y, Z so they will be redundant in $R_1 \bowtie R_2$.

So the final answer will depend on the minimum ^{no} of occurrence of in

between t_1 and t_2 .

For example:-

$R1$:	X	Y	Z
	1	2	3
	4	5	6

,

$R2$:	Y	Z	W
	2	3	6
	2	3	7
	5	6	8
	5	6	9

 $\Rightarrow R1 \bowtie R2$:

X	Y	Z	W
1	2	3	6
1	2	3	7
4	5	6	8
4	5	6	9

$\pi_{YZ} (R1 \bowtie R2)$:

Y	Z
2	3
2	3

ii) $R1 \bowtie_{R1.Y > R2.Y} R2$.

• MIN : 0

In the case where all of our Y values of $R1$ are smaller than $R2$'s Y.
then theta join can't return any tuple.

• MAX : $t_1 \times t_2$.

The complete opposite where all $R1.Y$ values are greater than $R2.Y$.
we get a cartesian product, i.e. all tuples.

For example :

our query :

$R1 \bowtie_{R1.Y > R2.Y} R2$.

$R1$:

X	Y	Z
2	1	9
3	1	8
4	1	7

$R2$:

Y	Z	W
100	1	4
100	2	5
100	3	6

$R1$:

X	Y	Z
100	1	4
100	2	5

$R2$:

Y	Z	W
2	1	9
3	1	8

$R1 \bowtie_{R1.Y > R2.Y} R2$:

$R1.X$	$R2.Y$	$R1.Z$	$R2.Y$	$R2.Z$	$R2.W$
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ie min : 0

$R1 \bowtie_{R1.Y > R2.Y} R2$:

$R1.X$	$R1.Y$	$R1.Z$	$R2.Y$	$R2.Z$	$R2.W$
100	1	4	2	1	9
100	1	4	3	1	8
100	2	5	2	1	9
100	2	5	3	1	8

max : $t_1 \times t_2$.

$$\text{iii) } \pi_x (R1 \bowtie \pi_{y,z} (R2)).$$

• MIN : Zero.

Considering the case where $R1, R2$ common attribute doesnot have any common value.

Then after selecting Y, Z from $R2$ if we do natural join with $R1$.
we will get no tuples, before selecting X .

• MAX : t_1 .

when we have $R1.Y, R1.Z = R2.Y, R2.Z$. we will have max no. of tuples, as after we select Y, Z from $R2$ we get common variable to ~~map~~ join with $R1$ in natural join, After selecting $R1.X$ we will get t_1 no. of tuples only.

for example, Our query : $\pi_x (R1 \bowtie \pi_{y,z} (R2)).$

$$R1: \begin{array}{|c|c|c|} \hline X & Y & Z \\ \hline 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \\ \hline \end{array}$$

$$R2: \begin{array}{|c|c|c|c|} \hline Y & Z & W \\ \hline 11 & 12 & 13 \\ 14 & 15 & 16 \\ 17 & 18 & 19 \\ \hline \end{array}$$

$$\pi_{Y,Z} (R2): \begin{array}{|c|c|} \hline Y & Z \\ \hline 11 & 12 \\ 14 & 15 \\ 17 & 18 \\ \hline \end{array}$$

$$R1 \bowtie \pi_{Y,Z} (R2): \begin{array}{|c|c|c|} \hline R1.X & R1.Y & R1.Z \\ \hline & & 0. \\ \hline \end{array}$$

$$R1: \begin{array}{|c|c|c|} \hline X & Y & Z \\ \hline 1 & 2 & 3 \\ 4 & 5 & 6 \\ \hline \end{array}$$

$$R2: \begin{array}{|c|c|c|} \hline Y & Z & W \\ \hline 2 & 3 & 10 \\ 5 & 6 & 11 \\ \hline \end{array}$$

$$\pi_{Y,Z} (R2): \begin{array}{|c|c|} \hline Y & Z \\ \hline 2 & 3 \\ 5 & 6 \\ \hline \end{array}$$

$$R1 \bowtie \pi_{Y,Z} (R2): \begin{array}{|c|c|c|} \hline R1.X & R1.Y & R1.Z \\ \hline 1 & 2 & 3 \\ 4 & 5 & 6 \\ \hline \end{array}$$

$$\pi_x (R1 \bowtie \pi_{Y,Z} (R2)): \begin{array}{|c|} \hline R1.X \\ \hline 1 \\ 4. \\ \hline \end{array}$$

$$iv) R1 - (\pi_{x,y}(R1) \bowtie (R1 \div \pi_x(R1)))$$

• MIN : Zero.

For our query to return minimum tuple we want, $(\pi_{x,y}(R1) \bowtie (R1 \div \pi_x(R1)))$

let that be $A = R1 \Rightarrow R1 - R1 = 0$.

For A to be $R1$, we want x, y, z attribute, we are getting x, y from $\pi_{x,y}(R1)$, so we need y, z from $R1 \div \pi_x(R1)$ to be common, for to make a good join.

Not only common, we want all tuples from $R1$, i.e. we want $R1$ to contain x value with every respective entries of y, z .

So in that way we will get $R1(y, z) \leftarrow R1 \div \pi_x(R1)$.

$$\bullet R1 \leftarrow \pi_{x,y}(R1) \bowtie R1(y, z)$$

$$\bullet \emptyset \leftarrow R1 - R1, \text{ i.e. } 0 \text{ tuples}$$

• MAX : t_1 .

Here we want $A = 0$.

So, we want natural join to fail, i.e. not even a single common tuple.

To get that we need $\emptyset \leftarrow R1 \div \pi_x(R1)$, i.e. not even a complete tuple in division criteria.

$$\text{ie } \bullet \emptyset \leftarrow R1 \div \pi_x(R1)$$

$$\bullet \emptyset \leftarrow \pi_{x,y}(R1) \bowtie \emptyset$$

$$\bullet R1 - \emptyset = R1, \text{ i.e. } t_1 \text{ no. of tuples.}$$

For example. Our Query: $R1 - (\pi_{x,y}(R1) \bowtie (R1 \div \pi_x(R1)))$.

$$R1: \begin{array}{c|ccc} & X & Y & Z \\ \hline & 1 & 9 & 9 \\ & 2 & 9 & 9 \end{array}$$

$$R1 \div \pi_x(R1) : \begin{array}{c|cc} & Y & Z \\ \hline & 9 & 9 \end{array}$$

$$\pi_{x,y}(R1) \bowtie \begin{array}{c|cc} & Y & Z \\ \hline & 9 & 9 \end{array}$$

$$R1 \begin{array}{c|cc} X & Y & \\ \hline 1 & 9 & \\ 2 & 9 & \end{array} \bowtie \begin{array}{c|cc} Y & Z & \\ \hline 9 & 9 & \end{array}$$

$$= R1: \begin{array}{c|ccc} & X & Y & Z \\ \hline & 1 & 9 & 9 \\ & 2 & 9 & 9 \end{array}$$

$$R1 - R1 = 0 : \text{min.}$$

$$R1: \begin{array}{c|ccc} & X & Y & Z \\ \hline & 1 & 2 & 3 \\ & 2 & 4 & 5 \end{array}$$

$$\pi_x(R1) = \begin{array}{c|c} R1 \cdot X & \\ \hline 1 & 2 \end{array}$$

$$R1 \div \pi_x(R1) = \begin{array}{c|ccc} & X & Y & Z \\ \hline & \emptyset & & \end{array}$$

$$\pi_{x,y}(R1) \bowtie \emptyset = \emptyset$$

$$R1 - \emptyset = R1 : t_1(\text{max}).$$

Q2. Given the relation $R_1(x, y, z)$ and $R_2(y, z, w)$, prove/disprove that the following relational algebra expression returns an empty relation as a result:

$$\pi_x(R_1 \bowtie (\pi_{y,z}(R_2) - \pi_{y,z}(R_1 \bowtie R_2)))$$

Proof:- Considering two cases.

1. When the values of common attribute are not common.

$$\text{ie Value-attr}(R_1) \cap \text{Value-attr}(R_2) = \emptyset$$

So, Natural Join of $R_1 \bowtie R_2 = \text{NULL}$ (no common attribute).

$$\pi_{y,z}(\emptyset) = \emptyset.$$

$$R_1 \bowtie \pi_{y,z}(R_2) = \emptyset \text{ NULL (no common attribute).}$$

$$\pi_x(\emptyset) = \text{NULL}.$$

2. When some tuples are common.

$$\text{ie Value-attr}(R_1) \cap \text{Value-attr}(R_2) = x > 0; [x \text{ is no. of common at valued attributes}].$$

So, Natural Join of $R_1 \bowtie R_2 = x \text{ no. of common tuples} = \{x\}$ (let).

$$\bullet \pi_{y,z}(R_1 \bowtie R_2) = \pi_{y,z}(x) = \text{all common joining valued attributes} = \{x'\} \text{ (let)}$$

$$\bullet \pi_{y,z}(R_2) - \{x'\} = \{y\} \text{ (ie all uncommon attributes of } R_2)$$

$$\bullet R_1 \bowtie \{y\} = \emptyset \text{ as}$$

$$\bullet \pi_x\{\emptyset\} = \text{NULL. OR Empty relation.}$$

As the above Query return NULL in all possible cases, we conclude YES, it returns an empty relation.